Department: Ambulatory	○Departmental ●Ambulatory-wide	On-Line Number: APP.DPC.015
Section: Policies & Procedures	**Effective Date:** 4/23/2013	
Title: Outpatient Treatment of Patients with Deep Vein Thrombosis (DVT)	**Revision Date:** 1/15, 11/15, 2/19	
Accountable Department or Committee: Urgent Care, Ambulatory Care Pharmacy and Inpatient Pharmacy, Pharmacy and Therapeutic Committee, Ambulatory Interdisciplinary Practice Committee	**Review Date:** 1/15, 11/15, 2/19	
Collaborating Subject Matter Expert(s): SCPMG Physicians from Department of Cardiology, Emergency Department, Urgent Care Department, Ambulatory Care Pharmacy and Inpatient Pharmacy	**Page Number**	1 of 7

REFERENCES:

- Arixtra Product information, Micromedix® Healthcare Series 2009 Arixtra [package insert]. Research Triangle Park, NC: GlaxoSmithKline; 2008
- Lovenox Product Information, Micromedex® Healthcare Series 2009 Lovenox® [package insert]. Bridgewater, NJ: Sanofi-Aventis; 2007

PURPOSE:

To manage outpatient treatment of Deep Vein Thrombosis (DVT) patients with Enoxaparin (Lovenox®) or Fondaparinux (Arixtra®) per protocol.

DEFINITIONS:

DVT – Deep Vein Thrombosis.

PROVISIONS/PROCEDURE:

INCLUSION AND EXCLUSION CRITERIA FOR OUTPATIENT ANTICOAGULATION SERVICE (OACS) PHARMACIST TO MANAGE OUTPATIENT TREATMENT of DVT PATIENTS with ENOXAPARIN (Lovenox®) or Fondaparinux (Arixtra®) PER PROTOCOL:

1. Inclusion Criteria:
    - Acute, upper or lower extremity DVT
    - Age 18 years and above
    - Currently on warfarin and INR is in sub-therapeutic range

2. Exclusion Criteria:
    - Active bleeding/high risk of hemorrhage
    - Hemodynamically unstable
    - Age greater than 75 years
    - History of stroke in past 6 weeks

- Non-cutaneous surgery in past 2 weeks
- Known Platelet count less than 75,000/mL. If unknown, leave blank.
- Signs or symptom pulmonary embolus
- Morbid obesity (absolute weight 150 kg = 330 lb.)
- Allergy to heparin or history of heparin induced thrombocytopenia (H.I.T.)
- Pregnancy
- Patient currently on Warfarin in therapeutic range. If sub-therapeutic range, the patient is eligible.

3. If the Patient meet the above criteria and if warfarin is prescribed, patient will be referred to OACS via Tapestry to start Outpatient Anticoagulation DVT Protocol. Length of treatment with warfarin for DVT should also be indicated in the referral.

4. If the patient has any of the above exclusion criteria and test results are positive, refer to Emergency Department (ED).

5. Do Not Use heparin if patient has Heparin Allergy or H.I.T.

SUBCUTANEOUS ANTICOAGULATANT DOSING GUIDELINES:

1. Enoxaparin (Lovenox®) – Preferred agent at Kaiser Permanente.

   A. Available in prefilled syringes: 30 mg, 40 mg, 60 mg, 80 mg, 100 mg, 120 mg, 150 mg

   B. PLEASE ROUND doses to **nearest tenth** unless patient has dexterity and/or visual conditions which may prevent patient from manipulating the syringe.

   C. In most cases (see below), preferred over Arixtra for renal insufficiency

Dosing Method	Dose (round to nearest syringe strength)	Note
Preferred (in CrCl greater than 30 ml/min)	1 mg/kg (actual body weight) Q 12 hr	Pts greater than 150 kg NOT eligible
Alternative (in CrCl greater than 30 ml/min)	1.5 mg/kg (actual body weight) Q24 hr	Pts greater than100 kg NOT eligible. Reserve ONLY for pts requiring home health administration
Severe Renal Impairment (in CrCl less than 30 ml/min)	1 mg/kg (actual body weight) Q 24 hr	Consider heparin in severe renal impairment and/or dialysis

*References: Lovenox Product Information, Micromedex® Healthcare Series 2009 Lovenox® [package insert]. Bridgewater, NJ: Sanofi-Aventis; 2007

2. Fondaparinux (Arixtra®) - Non-formulary agent at Kaiser Permanente

- Comes in prefilled syringes: 2.5mg, 5mg, 7.5mg, 10mg

A. **<u>DO NOT USE IF ANY OF THE FOLLOWING APPLY</u>:** Use Enoxaparin (Lovenox®)

- CrCl less than 80 ml/min (manufacturer guidelines for renal adjustment are NOT available)
- Age greater than 75 years old **AND** Weight less than 50 kg

*[Potentially GREATER BLEEDING RISK in above situations]

Dosing Table:

Actual Body Weight (kg)	Arixtra Dose
Less than 50	5 mg daily
50-100	7.5 mg daily
Greater than100	10 mg daily

*Arixtra Product information, Micromedix® Healthcare Series 2009 Arixtra [package insert]. Research Triangle Park, NC: GlaxoSmithKline; 2008

Appendix A – San Bernardino County Area Outpatient DVT Workflow

FONTANA AND ONTARIO MEDICAL CENTER OUTPATIENT DVT WORKFLOW Wells Score ≥ 3 = High Probability Wells Score 0-2 = Low Probability	
1. Physician to place **STAT lab orders** D-Dimer **PNL**: **85379B** (if low probability only) **(includes D-dimer, CBC, Creatinine, INR)**	2. Physician to place **STAT ultrasound order** to rule out DVT Right upper extremity: 93971E Left upper extremity: 93971D Right lower extremity: 93971C Left lower extremity: 93971H Bilateral lower extremity: 93970C
3.If Outside clinic/MOB, Physician contacts Urgent Care (UC) screening provider to give report/hand off. Screening provider to have patient scheduled as nurse visit. Clinic physician to inform patient of possible DVT diagnosis and follow up care planned (including lab work, possible ultrasound, and possible checking into Urgent Care).	
4. Patients from outlying clinics will be sent to Fontana or Ontario Medical Center. Clinic RN/LVN to instruct patient to check-in at the outpatient laboratory department located in Fontana Medical Center Medical Office Building 3 (MOB 3) or Ontario Medical Center Medical Office Building D (MOB D). Clinic RN/LVN to contact Radiology and schedule ultrasound exam at location patient sent to (Mon-Fri, 8am-5pm direct book, tieline 8-243-4040 or after hours and weekends 909-427-7650 or tieline 8-250-7650).	
5. Lab receptionist to instruct patient to wait at the lab waiting area and inform patient of estimated wait time if completing lab draw for D-Dimer (lower probability: see below)	
Higher Probability for DVT Wells Score ≥ 3	**Lower Probability for DVT Wells Score 0-2**
1. Patient to complete lab draw for CBC, Creatinine and INR 2. Clinic RN/LVN to instruct patient to go to Radiology for ultrasound directly after lab drawn. Patient **does not need to wait for lab** results.	1. Patient to complete lab draw for D-Dimer **PNL: 85379B (includes D-Dimer, CBC, Creatinine and INR)** 2. Clinic RN/LVN to instruct patient to wait for lab results

If Lab Results are Negative:	**If Lab Results are Positive:**
1. Lab representative to inform patient of lab results and send patient home	1. Lab representative to send patient to reception to check in for ultrasound
	2. Patient to wait for ultrasound result 3. If patient has left, unable to be found- Lab to notify UC of positive results. (Fontana UC X75750 or Ontario UC X 42412). 4. UC RN to attempt to contact patient and document.
If Ultrasound Result is Negative:	**If Ultrasound Result is Positive:**
1. Sonographer to send patient home	1. Sonographer to inform UC RN (Fontana UC X75750 or Ontario UC X 42412) of positive ultrasound and direct patient to reception to check in for UC Nurse visit.
	2. UC RN informs screening provider of test results.

3.UC Screening Physician to place Enoxaparin (Lovenox®) and any additional medication orders in HealthConnect.
4. If ordering Warfarin, UC provider will **send tapestry referral to Outpatient Anticoagulation Service (OACS)** to follow patient per Outpatient Anticoagulation DVT Protocol. Refer to DVT protocol for Enoxaparin (Lovenox®) dosing guidelines.
4. **On Fridays or Saturdays**, UC Physician will dose warfarin/ Enoxaparin (Lovenox®) through Monday or the next OACS business day and order INR/CBC for Mondays or the next OACS business day.

5. If the patient is sent home prematurely with a DVT and found to have positive ultrasound results, the radiologist will call UC physician to report final findings. UC Physician will:
a) Contact the patient and inform final DVT diagnosis.
b) If patient meets inclusion and exclusion criteria for outpatient DVT protocol, will advise patient to return to UC for medication and teaching ASAP.
c) If Warfarin prescribed, UC physician will send referral to OACS (referral not needed if patient is started on DOAC (Direct Oral Anticoagulants).
d) Advise patient to schedule follow up telephone appointment or appointment with PCP in 2-3 days.

First dose Enoxaparin and education

During FON/ONT UC Hours (Mon-Sun 9:00am-9:00pm, including Holidays and Weekends):

UC RN will:

1.Administer first dose Enoxaparin(Lovenox®) in UC department using Lovenox® from medication supply.
2. UC RN will provide patient education, handouts and instruct patient to pick up medication ordered at outpatient pharmacy. If Warfarin prescribed will also instruct patient to take first dose at home and continue other medication as prescribed.

For Warfarin prescription, RN will additionally contact:

1. Mon-Thur 9:00am to 5:00 pm and Fri 9:00 am to 3pm: **Notify Outpatient Anticoagulation Service (OACS) at 8-250-2129.** Provide patient's name, MRN, state referral is for new start Lovenox ® and warfarin, name of referring physician, and RN's call back number. OACS pharmacist will provide next day warfarin/Lovenox© dosing.

	<u>Fri after 3 PM, weekends and holidays</u>: Notify Fontana Medical Center Inpatient pharmacy at (909) 302-8339. Provide patient's name, MRN, state referral is for new start Lovenox ® and warfarin, name of referring physician, and RN's call back number. Inpatient pharmacist will provide next day/weekend warfarin/ Lovenox© dosing only if weekend/next day dosing is missing.
	2.UC RN will provide OACS phone number (909) 427-7364 for patient to call for any questions. 3.Close Encounter in HealthConnect within 24 hours
	During UC Afterhours (Mon-Sun 9:00pm – 9:00am): Refer/sign out patient to ED • Assigned attending physician to dose patient through Monday and order INR for Monday if patient is sent home on Friday • **IF FONTANA ED**: Assigned attending physician to call Fontana Inpatient pharmacy at (909) 302-8339 for dosing recommendation if needed • **IF ONTARIO ED**: Assigned attending physician to call Ontario Inpatient pharmacy at (909) 724-3500 for dosing recommendation if needed. • If patient does receive first dose of warfarin in ED, assigned attending physician and/or nurse needs to make sure patient's baseline INR is available. This is to meet Joint Commission National Patient Safety Goal requirement 3E associated with the use of anticoagulation therapy • Assigned attending physician to send tapestry referral to Outpatient Anticoagulation Service (OACS) to follow patient per Outpatient Anticoagulation DVT Protocol <u>and</u> ED clerk to notify OACS at **8-250-2129**. <u>Note</u>: A separate OACS Referral is required despite patient is being monitored by inpatient pharmacist per pharmacy protocol

Appendix B – Department Office Hours and Summary of Responsibility

FONTANA AND ONTARIO MEDICAL CENTERS

Department	Office Hours	Afterhours Coverage	Summary of Responsibility (refer to Appendix A for details)
Urgent Care RN	Mon – Sun 9:00am – 9:00pm, including weekends and holidays	**Emergency Department**	• Contact UC Physician • Administer First dose Lovenox • Provide medication Education • IF Warfarin ordered, contact OACS/Inpatient Pharmacy
UC Physician	Mon – Sun 9:00am – 9:00pm, including weekends and holidays	**Emergency Department**	• Order medications and provide instruction until OACS' next business day (if Warfarin) • Order CBC/INR on the next OACS business day (if Warfarin) • Send Tapestry Referral to OACS
Outpatient Anticoagulation Service (OACS) (8-250-2129)	M-F 9:00am - 5:00pm, except for Holidays (For Fridays, OACS accept DVT referral until 3PM)	**Fontana Inpatient Pharmacy** (909) 302-8339 **Ontario Inpatient Pharmacy** (909) 724-3500	• Provide Lovenox/warfarin instruction starting the next OACS business day (For example: if patient is diagnosed on Friday, OACS will contact patient on Monday)